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Q1 A dice is rolled once. Find probability of getting a number greater than 3.

Here

$n = 6$

Numbers greater than 3 = (4, 5, 6

Favoral outcomes = 3

So,

$$P = \frac{\text{favorable outcome}}{\text{sample space}} = \frac{3}{6} = \frac{1}{2}$$

$\therefore$ Probability of getting a number greater than 3 is $\frac{1}{2}$

Q2. A card is drawn from a deck of 52 cards. Find the probability of getting a face card.

Here,

Total cards $(n) = 52$

Face cards.

Jack = 4

Queen = 4

King = 4.

Total face cards $= 4+4+4$

$= 12$

Let A be a drawing a face card $n(A) = 12$.

Now $P = \frac{n(A)}{n(S)} = \frac{12}{52} = \frac{3}{13}$.

$\therefore$ Possibility of drawing a face card is $\frac{3}{13}$.

Q3. A bag contains 7 red, 5 blue and 8 green balls. One ball is selected. Find the probability that it is red.

Given

Red balls $= 7$

Blue balls $= 5$

Green balls $= 8$

Total ball $(n) = 7+5+8 = 20$

Let $A =$ Red ball

Now,

$$P = \frac{P(A)}{n} = \frac{7}{20}$$

$\therefore$ Probability that selected ball will be red is $\frac{7}{20}$.

Q4. A number is chosen from 1 to 20. Find the probability that it is divisible by 4.

Given.

Total numbers $n(S) = 20$

Numbers divisible by 4 are $(E) = \{4, 8, 12, 16, 20\}$

$n(E) = 5$

Now, $$P(E) = \frac{n(E)}{n(S)} = \frac{5}{20} = \frac{1}{4}$$

$\therefore$ Probability that number is divisible by 4 is $\frac{1}{4}$.

Q5. In a class of 50 students, 28 like Math, 22 like Science & 10 liked both. Find the probability that a randomly selected student like Math or Science.

Given

$n(S) = 50$ $\quad$ $n(Sc) = 22$

$n(M) = 28$ $\quad$ $n(M \cap Sc) = 10$

Now

$n(M \cup C) = n(M) + n(S_c) - n(M \cap C)$

$= 28 + 22 - 10$

$= 40$

Let, $\mathcal{E}$ be students who like Math or Science.

Now,

$$P(\mathcal{E}) = \frac{n(\mathcal{E})}{n(S)} = \frac{40}{50} = \frac{4}{5}$$

$\therefore$ Probability of randomly selected student liking Math or Science is $\frac{4}{5}$.

Q6. A card is drawn. Find the probability that it is a heart or a king.

Given

$n(S) = 52$

Let

A = Heart

B = King.

No. of hearts $= n(A) = 13$ $\quad$ $n(A \cap B) = 1$ (1 King of Hearts)

No. of king $= n(B) = 4$.

Now,

$$P(A \cup B) = \frac{n(A) + n(B) - (A \cap B)}{n(S)}$$

$$= \frac{13 + 4 - 1}{52}$$

$$= \frac{16}{52} = \frac{4}{13}$$

$\therefore$ Probability that it is a heart or a king is $\frac{4}{13}$.

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Q7. A dice is rolled. Find the probability of getting 2 or 5.

Given

$n(S) = 6$

Let A = getting 2

B = getting 5

Since, dice cannot show 2 & 5 at same time

$n(A) = 1$

$n(B) = 1$

$n(A \cap B) = 0$

Now, $P(A \cup B) = \frac{n(A) + n(B) - n(A \cap B)}{n(S)}$

$= \frac{1+1-0}{6} = \frac{2}{6} = \frac{1}{3}$

$\therefore$ Probability of getting 2 or 5 in a dice is $\frac{1}{3}$.

Q8. A card is drawn. Find the probability of drawing a queen or a jack.

Given.

$n(S) = 52$

Let A = Queen

B = Jack.

No. of queen, $n(A) = 4$

No. of jack, $n(B) = 4$

A card cannot be queen & jack, so $(A \cap B) = 0$

Now, $P(A \cup B) = \frac{n(A) + n(B) - n(A \cap B)}{n(S)}$

$= \frac{4+4-0}{52} = \frac{8}{52} = \frac{2}{13}$

$\therefore$ Probability of getting queen or jack is $\frac{2}{13}$

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Q9. 2 coins are tossed. Find probability of getting 2 heads.

Given.

Sample space, $S = \{HH, HT, TH, TT\}$

$n(S) = 4$

Let $E = \{HH\}$

No. of favorable outcome $n(E) = 1$

Now,

$$P(E) = \frac{n(E)}{n(S)} = \frac{1}{4}$$

$\therefore$ Probability of getting 2 heads is $\frac{1}{4}$.

Q10. A die is rolled & a coin is tossed. Find the probability of getting 4 & a head.

Given.

Die $= 6$ $\{1, 2, 3, 4, 5, 6\}$

Coin $= 2$ $\{H, T\}$

Total outcomes, $n(S) = 6 \times 2 = 12$

Sample space $= \{(1,H), (1,T), (2,H), (2,T), (3,H), (3,T), (4,H), (4,T),$
$(5,H), (5,T), (6,H), (6,T)\}$

Let $E = \{(4,H)\}$

No. of favourable outcome $= 1$

Now,

$$P(E) = \frac{n(E)}{n(S)} = \frac{1}{12}$$

$\therefore$ Probability of getting (4 & H) is $\frac{1}{12}$.

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Q.11. The probability that a machine works on a given day is 0.9. Another independent machine works with probability 0.8. Find the probability that both work.

Given

Let

A = First machine

B = Second machine.

Given, $P(A) = 0.9$

$P(B) = 0.8$

Now,

$$P(A \cap B) = P(A) \times P(B)$$
$$= 0.9 \times 0.8$$
$$= 0.72$$

$\therefore$ Probability that both machine works is 0.72.

Q.12 A student passes Physics with probability 0.7 & Chemistry with probability 0.6 independently. Find the probability that student passes at least 1 subject.

Given.

Let Physics $n(P) = 0.7$

Chemistry $n(C) = 0.6$

Now, $P(P \cap C) = n(P) \times n(C)$

$$= 0.7 \times 0.6$$
$$= 0.42$$

Then, $P(P \cap C) = A(P) + P(C) - P(P \cap C)$

$$= 0.7 + 0.6 - 0.42$$
$$= 1.3 - 0.42$$
$$= 0.88$$

$\therefore$ Probability of passing at least 1 subject is 0.88

Q13. A bag contains 5 white and 3 black balls. 2 balls are drawn without replacement. Find probability that both are white.

Given

white ball = 5

Black ball = 3

Let $n(W) = 5$ — Let $A = 1^{st}$ ball white

$n(B) = 3$ — $B = 2^{nd}$ white ball

$n(S) = 8$

Now, Probability that first ball was white.

$$P(A) = \frac{n(W)}{n(S)} = \frac{5}{8}$$

Now, Probability that second ball was white

$$P\left(\frac{B}{A}\right) = \frac{4}{7}$$

$$\text{Now, } (A \cap B) = P(A) \times P\left(\frac{B}{A}\right)$$

$$= \frac{5}{8} \times \frac{4}{7}$$

$$= \frac{20}{56} = \frac{5}{14}$$

∴ Probability that both balls were white is $\frac{5}{14}$

Q.14. A deck has 52 cards. 2 cards are drawn without replacement. Find the probability that both are aces.

Given,

$n(S) = 52$

$n(A) = 4$

Let A = First ace

B = Second ace.

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Now,

Probability that 1^{st} card is an ace, $P(A) = \frac{n(A)}{n(S)}$

$= \frac{4}{52} = \frac{1}{13}$

After one ace is removed

$n(A) = 3$

$n(S) = 51$

Again, Probability that 2^{nd} card is also an ace. $P\left(\frac{B}{A}\right) = \frac{3}{51}$

Now,

Probability that both card are ace, $P(A \cap B) = P(A) \times P\left(\frac{B}{A}\right)$

$= \frac{1}{13} \times \frac{\cancel{3}}{\cancel{51}\,17}$

∴ Probability that both card are ace $= \frac{1}{221}$

is ~~1~~/221.

15 A box contains 4 red & 6 blue balls. 2 balls are drawn without replace

Find ~~replacement~~ probability that the first is red & second is blue.

Given.

$n(S) = 4+6 = 10$

$n(R) = 4$

$n(B) = 6$

Let A = First ball is red

B = Second ball is blue

Now,

Probability that 1^{st} ball is red $P(A) = \frac{n(R)}{n(S)} = \frac{4}{10} = \frac{2}{5}$

After removing 1 ball

$n(S) = 9$

$n(B) = 6$

Probability that second ball is blue, $P(B) = P\left(\frac{B}{A}\right) = \frac{6}{9} = \frac{2}{3}$

Now.

Probability that 1st ball is red & second is blue, $P(A \cap B) = P(A) \times P\left(\frac{B}{A}\right)$

$= \frac{2}{5} \times \frac{2}{3}$

$= \frac{4}{15}$ //

Qno 16. 3 students are selected from a group of 5 boys & 4 girls. Find the probability that all selected students are girls.

Given.

No. of boys $n(B) = 5$

$n(G) = 4$

$n(S) = 5+4 = 9$

Now,

Probability that all selected students are girls.

Let A = 1st girl student

B = 2nd girl student

C = 3rd girl student

Soln 1st girl, $P(A) = \frac{n(G)}{n(S)} = \frac{4}{9}$

Now, no. of girls $n(G) = 3$

$n(S) = 8$

2nd girl, $P(B) = P\left(\frac{B}{A}\right) = \frac{3}{8}$

$n(G) = 2$

$n(S) = 7$

3rd ~~g~~ student is a girl, $P(C) = \frac{n(G)}{n(S)}$ = ~~4~~ $\frac{n(G)}{n(S)} = \frac{2}{7}$

$P\left(\frac{C}{A \cap B}\right)$

Now, Probability all students are girls

$P(A \cap B \cap C) = P(A) \times P\left(\frac{B}{A}\right) \times P\left(\frac{C}{A \cap B}\right)$

$= \frac{\cancel{4}}{\cancel{9}_3} \times \frac{\cancel{3}}{\cancel{8}_{\cancel{2}}} \times \frac{\cancel{2}}{7} = \frac{1}{21}$ //

17 A card ~~is~~ drawn is red. What is the probability that it is a king?

Given

$n(S) = 52$

$n(R) = 26$

No. of red king cards = 2

Let A = red king.

Now, $P(A) = \frac{n(A)}{n(S)} = \frac{2}{52} = \frac{1}{26}$ //

A dice shows an even number. What is the probability that it is greater than 4?

Given,

$n(S) = 3$ {1, 2, 3, 4, 5, 6}

$n(E) = 1$ {2, 4, 6}

Now, Probability that number is greater than 4 $P(A) = \frac{n(E)}{n(S)} = \frac{1}{3} = \frac{1}{3}$

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Q. 19 Factory A produces 70% of bulbs & 3% are defective. Factory B produces 30% & 5% are defective. A defective bulb is selected. Find the probability that it came from Factory B.

Given.

$n(A) = 70\% = 0.7$ $\quad P(A) = 0.7$ $\quad P(D/B) = 0.05$

$n(B) = 30\% = 0.3$ $\quad P(B) = 0.3$

$n(D) = 3\% = 0.03$ $\quad P(D/A) = 0.03$

Now, Using Baye's Theorem.

$$P\left(\frac{B}{D}\right) = \frac{P\left(\frac{D}{B}\right) \times P(B)}{P\left(\frac{D}{A}\right) \times P(A) + P\left(\frac{D}{B}\right) \times P(B)}$$

$$= \frac{0.05 \times 0.3}{0.03 \times 0.7 + 0.05 \times 0.3}$$

$$= \frac{0.015}{0.021 + 0.015} = \frac{0.015}{0.036} = 0.4167.$$

$\therefore$ Probability that defective bulb are from Factory B is 0.4167.

A disease occurs in 2% of people. A test gives positive for sick people with probability 96% & positive for healthy people with probability 8%. A person tests positive. Find the probability that the person actually has the disease.

Given.

$n(D) = 2\%$ $\quad 0.02$

Let D = Person has disease

P = Test result is positive.

Now, $P(D) = 0.02$

Healthy people $= P(D') = 1 - 0.02$ $\quad (100-2)$

$= 0.98$

Probability of positive test if sick $P\left(\frac{P}{D}\right) = 96\% = 0.96$.

Probability of positive test if healthy $P\left(\frac{P}{D'}\right) = 8\% = 0.08$

Using Baye's Theorem

$$P\left(\frac{D}{P}\right) = \frac{P\left(\frac{P}{D}\right) \times P(D)}{P\left(\frac{P}{D}\right) \times P(D) + P\left(\frac{P}{D'}\right) \times P(D')}$$

$$= \frac{0.96 \times 0.02}{0.96 \times 0.02 + 0.08 \times 0.98}$$

$$= \frac{0.0192}{0.0784 + 0.0976}$$

$$= \frac{0.0192}{0.0976} = 0.1967$$

$\therefore$ Probability that person has disease is 0.1967. //